

Cervical Shortening 03. No Prior Spontaneous Preterm Birth: Singleton

Companion reading handout for simultaneous listening.

Audio course on cervical shortening during pregnancy. Episode three. The asymptomatic singleton patient with no prior spontaneous preterm birth.

Opening frame

This is the cleanest population in the cervical shortening course, and that is why it deserves its own episode.

The patient is pregnant with a singleton. She has no prior spontaneous preterm birth. She is asymptomatic. No contractions. No bleeding. No leakage of fluid. No fever. No uterine tenderness. A standardized transvaginal ultrasound before twenty four weeks shows a short cervix.

This is the population most directly addressed by SMFM Consult Series number seventy.

Dominant model

The dominant model is this.

In the no-prior-spontaneous-preterm-birth singleton, isolated sonographic short cervix is usually a progesterone-and-surveillance problem, not a reflex cerclage problem.

That sentence prevents several common errors.

It prevents the error of ignoring the finding.

It prevents the error of placing cerclage in every short cervix.

It prevents the error of prescribing 17-OHPC because the word progesterone is in the resident's mind.

It prevents the error of prescribing bed rest because it feels intuitive.

Let us begin with the threshold.

SMFM recommends using a midtrimester transvaginal cervical length of twenty five millimeters or less to diagnose short cervix in a singleton gestation without a prior spontaneous preterm birth.

But treatment is not a single threshold.

For cervical length less than or equal to twenty millimeters diagnosed before twenty four weeks, SMFM recommends vaginal progesterone.

For cervical length twenty one to twenty five millimeters, SMFM recommends considering vaginal progesterone through shared decision-making.

That difference matters. It teaches that the cervix is a risk continuum, and that guideline strength changes as the cervix gets shorter and the evidence becomes more compelling.

Now, why vaginal progesterone?

Progesterone maintains pregnancy partly by supporting uterine quiescence and opposing cervical ripening pathways. Human labor is not usually triggered by a simple fall in serum progesterone. Instead, local progesterone action may become less effective through receptor changes and inflammatory signaling. Vaginal progesterone is biologically plausible because the cervix and lower uterine segment are part of the target tissue environment.

The evidence spine includes trials such as Fonseca and Hassan and the Romero individual patient data meta-analysis. Fonseca studied very short cervix and used two hundred milligrams of micronized vaginal progesterone nightly. Hassan's PREGNANT trial studied cervical length ten to twenty millimeters and used ninety milligrams of vaginal progesterone gel. These trials and the pooled evidence support reduction in early preterm birth and some neonatal morbidity outcomes in asymptomatic singleton short cervix.

SMFM notes that the most studied regimens are the ninety milligram gel and the two hundred milligram micronized capsule, but that evidence is insufficient to recommend one specific formulation over the other.

That is the honest way to teach the regimen.

Do not say, this is the dose because it is physiologically perfect. Say, these are the studied regimens, and local protocols may choose one.

Now the first trap.

Vaginal progesterone is not the same intervention as 17-alpha hydroxyprogesterone caproate.

17-OHPC was an intramuscular drug historically used for recurrent preterm birth prevention in selected patients. It is not recommended for treatment of short cervix. SMFM recommends against using 17-OHPC, including compounded formulations, for short cervix. ACOG's updated progestogen advisory also states that intramuscular 17-OHPC is not recommended for primary prevention of recurrent spontaneous preterm birth.

The resident reflex is this: do not substitute 17-OHPC for vaginal progesterone because both sound like progesterone.

Now the second trap: cerclage.

In the asymptomatic singleton patient without prior spontaneous preterm birth, SMFM recommends against cerclage placement for a sonographic cervical length between ten and twenty five millimeters when there is no cervical dilation.

This is a major point. Cerclage has a powerful intuitive appeal. The cervix is short; the stitch seems like it should help. But evidence does not support routine cerclage for this broad low-risk screened population.

The classic To trial is important here. Screened women with very short cervix, less than or equal to fifteen millimeters, did not show a broad significant benefit from cerclage in the general screened population. That does not mean cerclage never helps anyone. It means the phenotype matters.

Now the nuance.

What if the cervix is extremely short, less than ten millimeters?

SMFM does not turn this into a routine cerclage recommendation, but it allows consideration through shared decision-making in selected cases, based on lower-quality evidence. More importantly, very short cervix should prompt consideration of cervical examination because dilation may be present. If dilation is present, you are no longer managing isolated sonographic shortening. You are in the physical-exam-indicated cerclage world.

So in a no-prior-spontaneous-preterm-birth singleton with cervical length eighteen millimeters and a closed cervix, the answer is vaginal progesterone.

In the same patient with no measurable cervix or a cervix of eight millimeters, the answer is not simply more anxiety. It is: evaluate for dilation, discuss the uncertainty, and involve MFM level judgment.

Now the third trap: pessary.

Pessary is mechanically attractive. It may alter the uterocervical angle, redistribute uterine pressure, reduce opening of the internal os, or protect the mucus plug. The problem is that plausible mechanism has not translated into consistent trial benefit.

Goya's PECEP trial was strongly positive. Nicolaides' larger singleton pessary trial was negative. SMFM recommends against cervical pessary for prevention of preterm birth in singleton short cervix. SuPPoRT later compared cerclage, pessary, and progesterone in a high-risk short-cervix population and did not show clear superiority of one strategy, but it does not overturn the SMFM no-prior singleton framework.

The lesson is trial literacy. A dramatic early trial is not the end of evidence when later larger or different trials do not replicate the effect.

Now the fourth trap: bed rest.

Do not prescribe bed rest or broad activity restriction as evidence-based prevention. It does not treat isolated short cervix. It can cause harm: thromboembolism risk, bone loss, deconditioning, psychosocial stress, family disruption, and financial harm. SMFM Consult number fifty, reaffirmed in twenty twenty five and endorsed by ACOG, recommends against routine activity restriction for shortened cervix or preterm birth risk. ACOG Committee Opinion eight zero four also states that activity restriction should not be prescribed routinely to reduce preterm birth.

The resident should not say, do whatever you want, and disappear. The better counseling is targeted.

Avoid activities that trigger contractions, bleeding, leakage, pain, or pressure. Seek care if those symptoms occur. We may individualize work or exercise recommendations. But strict bed rest is not a treatment for your cervix.

Bedside algorithm

Now let us build the actual bedside algorithm.

Clinical cases

A patient is twenty one weeks. Singleton. No prior spontaneous preterm birth. Anatomy scan suggests a short cervix.

Step one: confirm a technically adequate transvaginal ultrasound measurement.

Step two: confirm she is asymptomatic. Ask about contractions, pelvic pressure, bleeding, leakage of fluid, fever, discharge, pain, and uterine tenderness.

Step three: clarify history. No prior spontaneous preterm birth means no prior spontaneous labor or PPROM leading to preterm delivery. Still ask about midtrimester loss, cervical surgery, uterine anomalies, and prior cerclage, because the story may not be as low-risk as the label suggests.

Step four: interpret the cervical length.

If it is more than twenty five millimeters, she does not meet the standard short cervix threshold, but risk is still a continuum.

If it is twenty one to twenty five millimeters before twenty four weeks, discuss vaginal progesterone using shared decision-making.

If it is less than or equal to twenty millimeters before twenty four weeks, recommend vaginal progesterone.

If it is extremely short, especially less than ten or eleven millimeters or no measurable cervix, strongly consider sterile cervical exam to assess dilation and

discuss MFM involvement.

Step five: do not add unsupported interventions by reflex. No 17-OHPC. No routine cerclage for ten to twenty five millimeters without dilation. No routine pessary. No bed rest.

Now counseling.

The patient hears, short cervix, and may think delivery is imminent. The resident must be clear without being falsely reassuring.

Patient counseling

Try this:

Your cervix is shorter than expected for this point in pregnancy. That increases the risk of preterm birth, but it does not mean preterm birth is certain. In patients like you, the treatment with the best evidence is vaginal progesterone, especially when the cervix is twenty millimeters or shorter before twenty four weeks. We also need to make sure there is no dilation, bleeding, leaking fluid, infection, or contractions, because those would change the problem.

That counseling line gives risk, treatment, and the reason for re-triage if symptoms appear.

Now a case.

Twenty two weeks. TV cervical length nineteen millimeters. No prior birth. No contractions. No leakage. No bleeding. Cervix closed on exam because the clinician checked due to very short length.

What is the resident reflex?

Recommend vaginal progesterone. Explain the risk. Do not place cerclage by default. Do not prescribe 17-OHPC. Do not recommend bed rest. Give clear return precautions.

Second case.

Twenty two weeks. TV cervical length twenty four millimeters. No prior preterm birth.

This is not nothing. It is short by the twenty five millimeter definition. But SMFM places this in a shared decision-making zone for vaginal progesterone. The conversation should include the uncertainty, the patient's preferences, the low harm profile, and local protocol.

Third case.

Twenty two weeks. Cervical length seven millimeters. No prior spontaneous preterm birth. Asymptomatic.

Do not manage this as a routine nineteen millimeter cervix. Extremely short cervix raises concern for occult dilation. The next step is careful evaluation, usually including sterile speculum or digital examination depending on local practice and MFM input. If dilation is found, the lecture changes.

Final synthesis

Let us close.

In this population, the clean resident reflex is:

Confirm TV ultrasound. Rule out symptoms and dilation. Use vaginal progesterone for the right threshold. Do not reflexively add cerclage, pessary, 17-OHPC, or bed rest.

The millimeter matters. But the phenotype matters more.